

Math 526, F08
Quiz 1

Name: *Answer*
SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given $\mathbf{v} = (1, 1)$ and $\mathbf{w} = (1, 5)$.

(a) Find the *cosine* of the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

Solution: $\|\mathbf{v}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$$\|\mathbf{w}\| = \sqrt{1^2 + 5^2} = \sqrt{26}$$

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{1 \cdot 1 + 1 \cdot 5}{\sqrt{2} \sqrt{26}} = \frac{6}{2\sqrt{13}} = \frac{3}{\sqrt{13}}$$

(b) Choose a number c so that $\mathbf{w} + c\mathbf{v}$ is perpendicular to $\mathbf{v}$.

Solution: $\mathbf{w} + c\mathbf{v} \perp \mathbf{v} \Leftrightarrow (\mathbf{w} + c\mathbf{v}) \cdot \mathbf{v} = 0$

$$\mathbf{w} \cdot \mathbf{v} + c\mathbf{v} \cdot \mathbf{v} = 0$$

$$c = -\frac{\mathbf{w} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}$$

$$= -\frac{6}{2}$$

$$= -3$$

Problem 2. (10 points) Compute Ax for the matrix A and the vector x

$$A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

by the following ways:

(a) Using dot products of the rows of A with the column vector x .

Solution:

$$\begin{aligned} Ax &= \begin{bmatrix} (1, 2, 4) \cdot (2, 2, 3) \\ (-2, 3, 1) \cdot (2, 2, 3) \\ (-4, 1, 2) \cdot (2, 2, 3) \end{bmatrix} \\ &= \begin{bmatrix} 1 \cdot 2 + 2 \cdot 2 + 4 \cdot 3 \\ (-2) \cdot 2 + 3 \cdot 2 + 1 \cdot 3 \\ (-4) \cdot 2 + 1 \cdot 2 + 2 \cdot 3 \end{bmatrix} \\ &= \begin{bmatrix} 18 \\ 5 \\ 0 \end{bmatrix} \end{aligned}$$

(b) Using linear combination of the columns of A with components of x as coefficients.

Solution:

$$\begin{aligned} Ax &= 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -4 \\ -8 \end{bmatrix} + \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix} + \begin{bmatrix} 12 \\ 3 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 2+4+12 \\ -4+6+3 \\ -8+2+6 \end{bmatrix} \\ &= \begin{bmatrix} 18 \\ 5 \\ 0 \end{bmatrix} \end{aligned}$$